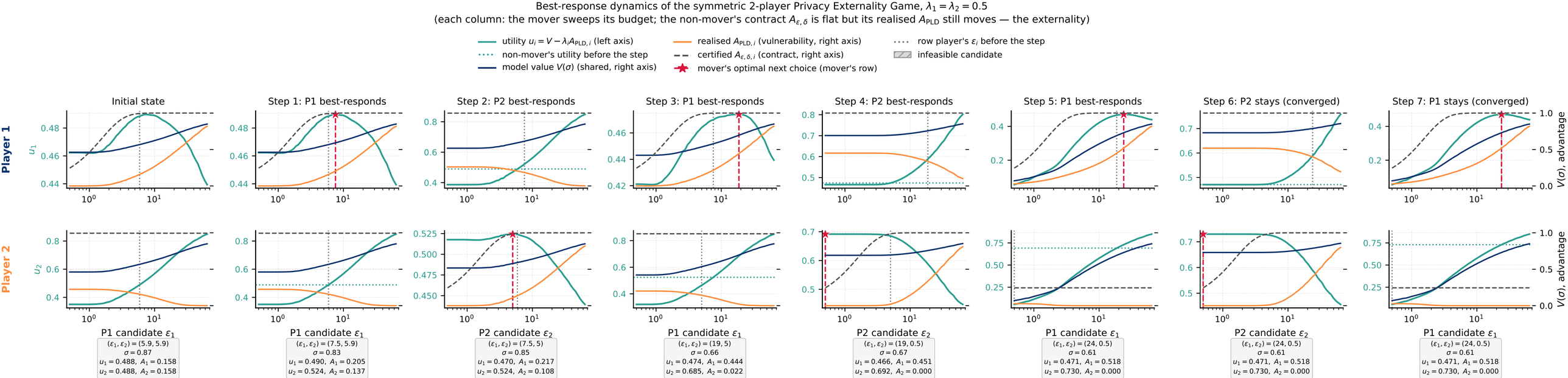

Best-response dynamics of the symmetric 2-player Privacy Externality Game, $\lambda_1 = \lambda_2 = 0.5$
(each column: the mover sweeps its budget; the non-mover's contract $A_{\varepsilon,\delta}$ is flat but its realised A_{PLD} still moves — the externality)
utility $u_i = V - \lambda_i A_{\mathrm{PLD},i}$ (left axis)
non-mover's utility before the step
model value $V(\sigma)$ (shared, right axis)
realised $A_{\mathrm{PLD},i}$ (vulnerability, right axis)
certified $A_{\varepsilon,\delta,i}$ (contract, right axis)
mover's optimal next choice (mover's row)
row player's ε_i before the step
infeasible candidate
Player 1
Player 2
u_1
u_2
$V(\sigma)$, advantage
Initial state
Step 1: P1 best-responds
Step 2: P2 best-responds
Step 3: P1 best-responds
Step 4: P2 best-responds
Step 5: P1 best-responds
Step 6: P2 stays (converged)
Step 7: P1 stays (converged)
P1 candidate ε_1
P2 candidate ε_2
$(\varepsilon_1, \varepsilon_2) = (5.9, 5.9)$
$\sigma = 0.87$
$u_1 = 0.488$, $A_1 = 0.158$
$u_2 = 0.488$, $A_2 = 0.158$
$(\varepsilon_1, \varepsilon_2) = (7.5, 5.9)$
$\sigma = 0.83$
$u_1 = 0.490$, $A_1 = 0.205$
$u_2 = 0.524$, $A_2 = 0.137$
$(\varepsilon_1, \varepsilon_2) = (7.5, 5)$
$\sigma = 0.85$
$u_1 = 0.470$, $A_1 = 0.217$
$u_2 = 0.524$, $A_2 = 0.108$
$(\varepsilon_1, \varepsilon_2) = (19, 5)$
$\sigma = 0.66$
$u_1 = 0.474$, $A_1 = 0.444$
$u_2 = 0.685$, $A_2 = 0.022$
$(\varepsilon_1, \varepsilon_2) = (19, 0.5)$
$\sigma = 0.67$
$u_1 = 0.466$, $A_1 = 0.451$
$u_2 = 0.692$, $A_2 = 0.000$
$(\varepsilon_1, \varepsilon_2) = (24, 0.5)$
$\sigma = 0.61$
$u_1 = 0.471$, $A_1 = 0.518$
$u_2 = 0.730$, $A_2 = 0.000$
$(\varepsilon_1, \varepsilon_2) = (24, 0.5)$
$\sigma = 0.61$
$u_1 = 0.471$, $A_1 = 0.518$
$u_2 = 0.730$, $A_2 = 0.000$
$(\varepsilon_1, \varepsilon_2) = (24, 0.5)$
$\sigma = 0.61$
$u_1 = 0.471$, $A_1 = 0.518$
$u_2 = 0.730$, $A_2 = 0.000$